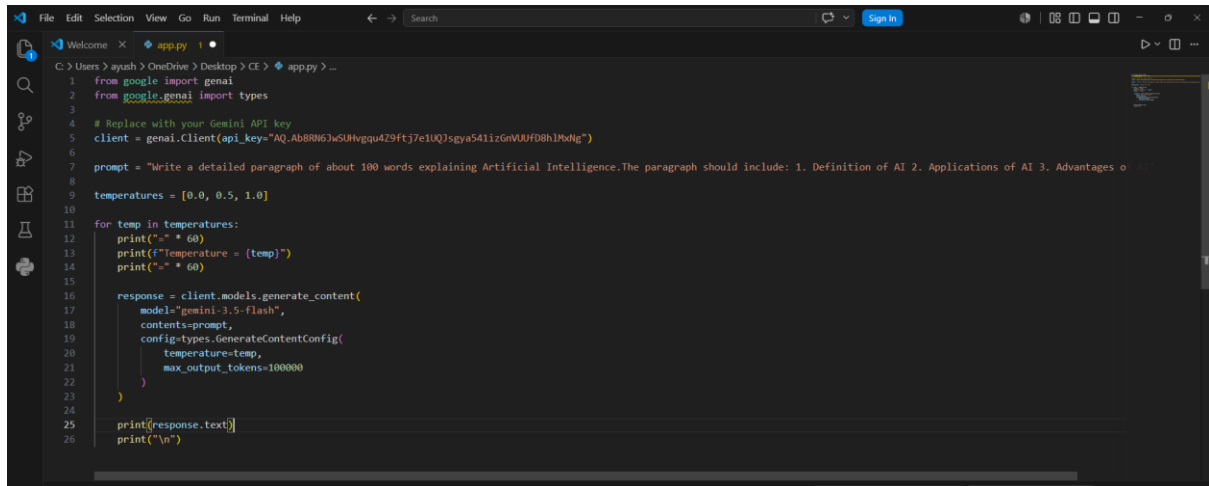


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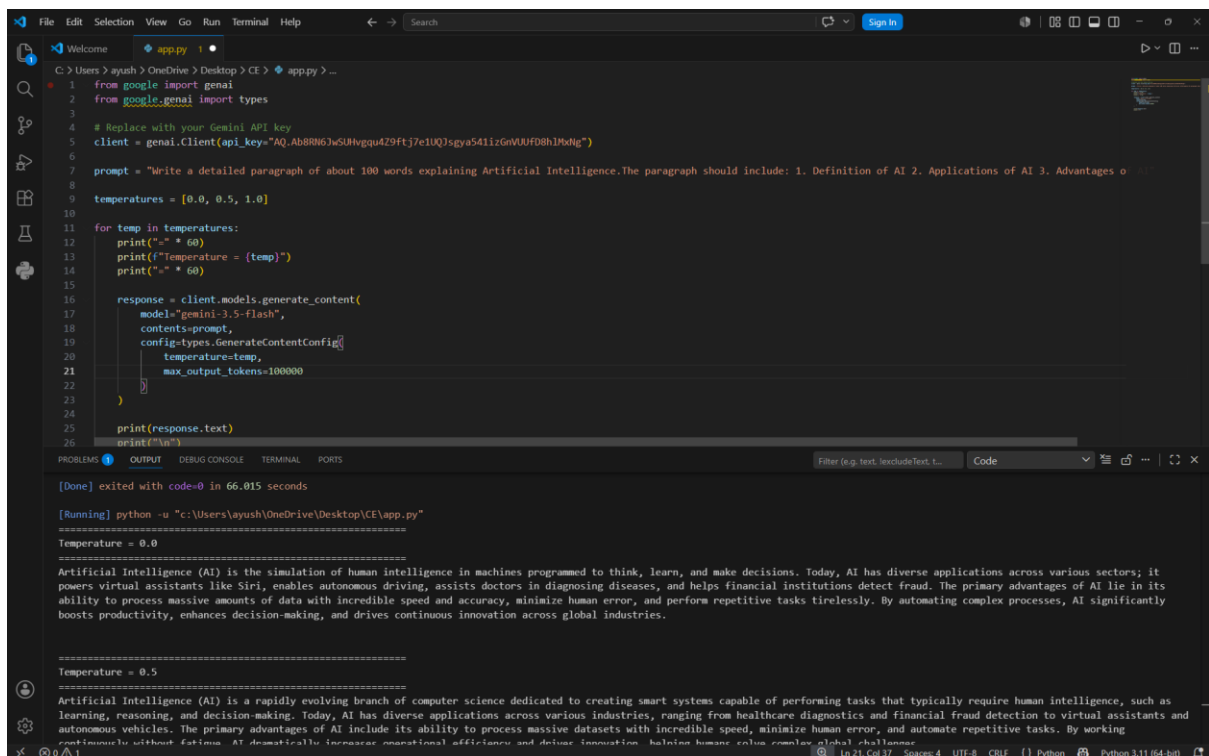
Task : Write a program to generate prompt responses using different temperature using any gpt model.

CODE :



```
1 from google import genai
2 from google.genai import types
3
4 # Replace with your Gemini API key
5 client = genai.Client(api_key="AQ-Ab8R96JwSUHvggu4Z9ftj7e1UQJsgya541izGwUUF08hJHdNg")
6
7 prompt = "Write a detailed paragraph of about 100 words explaining Artificial Intelligence. The paragraph should include: 1. Definition of AI 2. Applications of AI 3. Advantages of AI"
8
9 temperatures = [0.0, 0.5, 1.0]
10
11 for temp in temperatures:
12     print("-" * 60)
13     print(f"Temperature = {temp}")
14     print("-" * 60)
15
16     response = client.models.generate_content(
17         model="gemini-3.5-flash",
18         contents=prompt,
19         config=types.GenerateContentConfig(
20             temperature=temp,
21             max_output_tokens=100000
22         )
23     )
24
25     print(response.text)
26     print("\n")
```

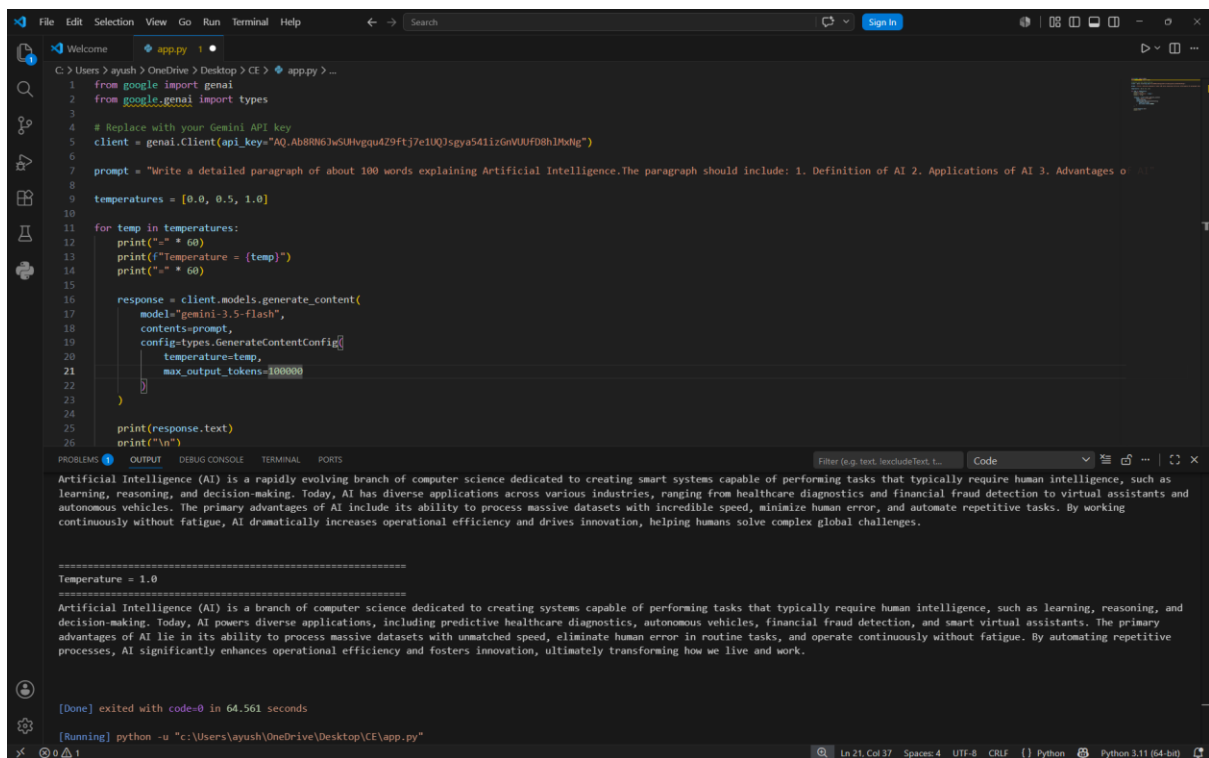
OUTPUT :



```
[Done] exited with code=0 in 66.015 seconds

[Running] python -u "c:\Users\ayush\OneDrive\Desktop\CE\app.py"
=====
Temperature = 0.0
=====
Artificial Intelligence (AI) is the simulation of human intelligence in machines programmed to think, learn, and make decisions. Today, AI has diverse applications across various sectors; it powers virtual assistants like Siri, enables autonomous driving, assists doctors in diagnosing diseases, and helps financial institutions detect fraud. The primary advantages of AI lie in its ability to process massive amounts of data with incredible speed and accuracy, minimize human error, and perform repetitive tasks tirelessly. By automating complex processes, AI significantly boosts productivity, enhances decision-making, and drives continuous innovation across global industries.

Temperature = 0.5
=====
Artificial Intelligence (AI) is a rapidly evolving branch of computer science dedicated to creating smart systems capable of performing tasks that typically require human intelligence, such as learning, reasoning, and decision-making. Today, AI has diverse applications across various industries, ranging from healthcare diagnostics and financial fraud detection to virtual assistants and autonomous vehicles. The primary advantages of AI include its ability to process massive datasets with incredible speed, minimize human error, and automate repetitive tasks. By working continuously without fatigue, AI dramatically increases operational efficiency and drives innovation, balancing human values against complex global challenges.
```



```
1 from google import genai
2 from google.generativeai import types
3
4 # Replace with your Gemini API key
5 client = genai.Client(api_key="AQ.Ab8R06JwSUHvgu4Z9ftj7e1UQ)sgya541izGnVUUF08hJMdHg")
6
7 prompt = "Write a detailed paragraph of about 180 words explaining Artificial Intelligence. The paragraph should include: 1. Definition of AI 2. Applications of AI 3. Advantages of AI"
8
9 temperatures = [0.0, 0.5, 1.0]
10
11 for temp in temperatures:
12     print("-" * 60)
13     print(f"Temperature = {temp}")
14     print("-" * 60)
15
16     response = client.models.generate_content(
17         model="gemini-1.5-flash",
18         contents=prompt,
19         config=types.GenerateContentConfig(
20             temperature=temp,
21             max_output_tokens=100000
22         )
23     )
24     print(response.text)
25     print("\n")
26
```

Artificial Intelligence (AI) is a rapidly evolving branch of computer science dedicated to creating smart systems capable of performing tasks that typically require human intelligence, such as learning, reasoning, and decision-making. Today, AI has diverse applications across various industries, ranging from healthcare diagnostics and financial fraud detection to virtual assistants and autonomous vehicles. The primary advantages of AI include its ability to process massive datasets with incredible speed, minimize human error, and automate repetitive tasks. By working continuously without fatigue, AI dramatically increases operational efficiency and drives innovation, helping humans solve complex global challenges.

Temperature = 1.0

Artificial Intelligence (AI) is a branch of computer science dedicated to creating systems capable of performing tasks that typically require human intelligence, such as learning, reasoning, and decision-making. Today, AI powers diverse applications, including predictive healthcare diagnostics, autonomous vehicles, financial fraud detection, and smart virtual assistants. The primary advantages of AI lie in its ability to process massive datasets with unmatched speed, eliminate human error in routine tasks, and operate continuously without fatigue. By automating repetitive processes, AI significantly enhances operational efficiency and fosters innovation, ultimately transforming how we live and work.

[Done] exited with code=0 in 64.561 seconds

[Running] python -u "c:\Users\ayush\OneDrive\Desktop\CE\app.py"

Explanation of the Code

1. Import Required Libraries

```
from google import genai
```

```
from google.generativeai import types
```

Explanation:

- `google.generativeai` is the Python SDK used to access Google's Gemini AI models.
- `types` provides configuration options such as temperature and maximum output tokens.

2. Create the Gemini Client

```
client = genai.Client(api_key="YOUR_GEMINI_API_KEY")
```

Explanation:

- A client object is created to establish a connection with the Gemini API.
- The API key authenticates the user and allows access to the Gemini model.

3. Define the Prompt

```
prompt = "Explain Artificial Intelligence in about 100 words."
```

Explanation:

- The prompt is the input text sent to the AI model.
 - The model generates a response based on this instruction.
-

4. Create a List of Temperature Values

```
temperatures = [0.0, 0.5, 1.0]
```

Explanation:

- A list of different temperature values is created.
 - Each value controls the creativity of the AI-generated response.
-

5. Iterate Through Each Temperature

```
for temp in temperatures:
```

Explanation:

- The loop executes once for each temperature value.
 - This allows the same prompt to be tested with different creativity levels.
-

6. Display the Current Temperature

```
print("=" * 60)
```

```
print(f"Temperature = {temp}")
```

```
print("=" * 60)
```

Explanation:

- Prints a heading showing which temperature is currently being used.
 - Improves readability of the output.
-

7. Generate the AI Response

```
response = client.models.generate_content(
```

```
model="gemini-3.5-flash",
contents=prompt,
config=types.GenerateContentConfig(
    temperature=temp,
    max_output_tokens=300
)
)
```

Explanation:

- Sends the prompt to the Gemini model.
 - temperature=temp controls the randomness of the response.
 - max_output_tokens=300 limits the maximum length of the generated text.
 - The generated response is stored in the response variable.
-

8. Print the Generated Response

```
print(response.text)
```

Explanation:

- Displays the AI-generated response for the selected temperature.
-

Explanation of the Output

The program generates responses for the same prompt using three different temperature values.

Output 1

Temperature = 0.0

Artificial Intelligence (AI)

Explanation

- Temperature **0.0** produces the most deterministic output.
- The model attempts to generate a factual and consistent response.

- In your execution, the response was very short due to the model or generation configuration.

Output 2

Temperature = 0.5

Artificial Intelligence (AI) is

Explanation

- Temperature **0.5** balances accuracy and creativity.
- The model starts generating a more descriptive sentence.
- The response is still short because of the current model behavior or request configuration.

Output 3

Temperature = 1.0

Artificial Intelligence (AI) refers

Explanation

- Temperature **1.0** allows the model to generate more diverse and creative responses.
- The model begins the explanation differently compared to the previous outputs.
- Although the response is truncated in your run, it demonstrates that changing the temperature changes how the model starts generating text.

Observation Table

Temperature	Output Generated	Interpretation
0.0	Artificial Intelligence (AI)	Most deterministic and factual output.
0.5	Artificial Intelligence (AI) is	Slightly more descriptive than temperature 0.0.
1.0	Artificial Intelligence (AI) refers	More creative and varied beginning of the response.

Result

The program successfully generated responses for the same prompt using different temperature values. The experiment demonstrates that the **temperature parameter influences the randomness and creativity of the generated text**. Lower temperatures produce more deterministic responses, while higher temperatures encourage more diverse and creative text generation. Although the responses in this execution were shorter than expected, they still illustrate the effect of changing the temperature parameter.